

Ramratan Padhy

+91 7735334802 | ramratanpadhy59@gmail.com
github.com/consolelogram | linkedin.com/in/ramratan-ntlap

EDUCATION

SRM University AP Amaravati, Andhra Pradesh
Bachelor of Technology in Computer Science Engineering 2025 – Present
• **Year 1** | SGPA: 9.0/10.0

EXPERIENCE

Next Tech Lab AP Amaravati, Andhra Pradesh
Associate Oct 2025 – Present
• Research associate at a QS ReImagine awarded student society.
• Collaborating with a research collective to explore low-level implementations of emerging technologies.
• Contributing to peer-to-peer technical workshops on systems programming and AI architecture.

Notecove (Stealth Startup) Remote
Technical Content Writer May 2025 – Aug 2025
• Authored 30+ comprehensive technical articles and guides on emerging software trends and AI ethics.
• Synthesized complex engineering concepts into accessible documentation for a specialized audience.
• Collaborated with the founding team to define the technical voice and content strategy for the platform.

Hush (Startup) Remote
ML Intern March 2026 – Present
• Worked with image models to discern user profiles to keep the app authenticity.
• Collaborated with the founding team to define and enhance user experience.

PROJECTS

Grand Tribunal | *Python, Modal, Nemotron, VoiceCPM* Completed
• Shipped an Ace Attorney-style turn-based AI debate game utilizing voice input for an immersive, hands-free player experience. Trained and deployed two specialized Nemotron models on character personality datasets to serve as dynamic opponents and logical judges, interfaced through a custom Gradio frontend.

Argus | *Python, Gemma 4, IoT Simulation* Completed
• Simulated a drone swarm that collectively maps any terrain using Gemma 4 31B vision via Cerebras.
• Engineered autonomous agent drones sharing a collective memory that aggregates into a unified semantic map displayed on a CesiumJS 3D globe.
• Used Gemma 4 inference to classify areas on the terrain, map traffic congregation spots
• Worked with FastAPI to serve as the backend for the simulation

High-Performance Editorial Portfolio | *HTML, CSS, Web Optimization* Completed
• Architected a zero-dependency, single-file portfolio site optimized for instant edge delivery, achieving sub-300ms load times and a perfect 100/100 Lighthouse performance score.
• Engineered a completely static, JavaScript-free architecture that inherently eliminates third-party vulnerabilities, maximizing security while ensuring optimal crawlability for modern AEO and SEO bots.

AXtheR (Windows) | *C, Windows API, Wintun, Cryptography* Completed
• Developed a high-performance, zero-dependency, easy to run UDP VPN for Windows by integrating the Wintun kernel driver and Winsock2.
• Engineered a zero-heap allocation data path with multi-threaded asynchronous I/O and dynamic Windows routing table manipulation.
• Implemented a self-contained, RFC-compliant ChaCha20-Poly1305 cryptographic engine from scratch in C with constant-time MAC verification.

Independent Research on BitNet-1.58b | *Python, Machine Learning, Mathematics* Completed
• Conducting novel research and experimentation on BitNet-1.58b by altering its context length.

TECHNICAL SKILLS

Languages: Python, C, LaTeX, Godot (GDScript)
Developer Tools: APIs, Vector Databases, PyTorch, Scikit-Learn, TensorFlow, Windows API, Antigravity
Core Concepts: Data Structures, Transformer Architectures, LLaMA Architecture, RAG, Technical Documentation, Mechanistic Interpretability, Model Evaluation
Soft Skills: Technical Communication, Research Synthesis, Problem Solving, Teamwork